

ATM Network Security Protocol

Module 29

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Video: <https://www.youtube.com/watch?v=JdRurgWuHvQ>

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1 second [Music] Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture we will study about ATM network security protocol so what is an ATM asynchronous transfer mode ATM is a switching technique used by telecommunication networks that uses asynchronous time division multiplexing to encode data into small fixed sized cells this is different from Ethernet or Internet which use variable packet sizes for data or frames ATM is the core protocol used over the synchronous Optical Network Sonet backbone of the ISDN asynchronous transfer mode was designed with sales in mind this is because voice data is converted to packets and is forced to share a network with burst data passing through the same medium so no matter how small the voice

packets are they always encounter full size data packets and could experience maximum queuing delays this is why all data packets should be of the same size the fixed Sal structure of of ATM means it can be easily switched by Hardware without the delays introduced by rooted frames and software switching this is why some people believe that ATM is the key to the internet bandwidth problem ATM creates fixed Roots between two points before data transfer begins which differs from tcpip where data is divided into packets Each of which takes a different route to get to its destination this makes it easier to build data usage however an ATM network is less adaptable to a sudden Network traffic search the ATM provides data link layer services that run on the OSI

layer one physical links it functions much like small packet switched and circuit switch networks which makes it idal for real time low latency data such as viip and video as well as for high throughput data traffic like file transfers a virtual circuit or connection must be established before the two end points can actually exchange data ATM Services generally have four different bit rate choices available bit rate provides a guaranteed minimum capacity but data can be bursted to higher capacities when Network traffic is minimal constant bit rate specifies a fixed bit rate so that data is sent in a steady stream this is analogous to a leased line unspecified bit rate doesn't guarantee any throughput level and is used for applications such as file transfers that can tolerate delays variable bit rate

vbr provides a specified throughput but data is not sent evenly this makes it a even popular choice for voice and video conferencing the structure of an ATM cell an ATM cell consists of a 5 byte header and a 48 by payload ATM defines two different cell formats as shown in the figure uni user network interface and nni that is network network interface most ATM links use uni cell format GFC that is generic flow control four bits and default is 40 bits VPI that is virtual path identifier 8 Bits uni or 12 Bits n ni I VCI virtual Channel identifier 16 bits PT P type that is three bits PT bit three Network management cell if zero user data cell and the following apply PT bit 2 explicit forward conen indication efc1 1 is equal to network congestion

experienced PT bit 1 On's bit ATM user to user AA bit used by aal5 to indicate packet boundaries CLP that is cell loss priority 1 bit HC that is header error control which is 8bit CRC polinomial is equal to $x^8 + x^2 + 1$ ATM uses the PT field to designate various special kinds of sales for operations Administration and management that is om purposes and to delineate packet boundaries in some ATM adaptation layers A A if the most significant bit of the PT field is zero this is a user data cell and the other two bits are used to indicate Network congestion and as a general purpose header bit available for ATM adaptation layers if the MSB bit of the PT bit is one this is a management cell and the other two bits indicate the type that is

Network management segment Network management end to end resource management and reserved for future use several ATM link protocols use the HC field to drive a CRC based framing algorithm which allows locating the ATM cells with no overhead beyond what is otherwise needed for header protection the 8bit CRC is used to correct single bit header errors and detect multi-bit header errors when multi-bit header errors are detected the current and subsequent sales are dropped until a cell with no header errors is found a un cell reserves the GFC field for a local flow controls sub multiplexing system between users this was intended to allow several terminals to share a single network connection in

the same way that two integrated Services digital Network ISDN phones can share a single basic rate ISDN connection all four GFC bits must be Zero by default the nni cell format replicates the uni format almost exactly except that the 4bit GFC field is reallocated to the V field extending the VPI to 12 bits thus the single nni ATM interconnection is capable of addressing almost 2 to the power 16 VPS of up to almost 2 to the power 16 VCS each in practice some of the VP and VC numbers are reserved ATM network interfaces following are the ATM network interfaces and are shown in the figure user to network interface public network interface private network interface second Network to node interface which includes private nni public

nni data exchange interface dxi between packet routers and ATM digital service units DSU the ATM adaption layer how to break application messages to sales the ATM layer transmission switching 8 minutes, 1 second reception congestion control buffer management cell header generation removal at source or destination reset connection identifiers for the next hope at

switch sell address translation sequential delivery the ATM Forum specifications data confidentiality the function of confidentiality is to protect precious business data from unauthorized persons confidentiality part of network security makes sure that the data is available only to the intended and authorized persons data integrity and authentication this goal means maintaining and assuring the accuracy

and consistency of data the function of Integrity is to make sure that the data is reliable and is not changed by unauthorized persons AAL frames are protected Ed by appending a cryptographic check sum to a frames possibility is given to provide reordering protection by introducing a sequence number into AAL frames before calculating the cryptographic check sum signaling protection any signaling messages may be authenticated and integrity protected by introducing a digital signature into an siie especially if anme protocol is employed protection is offered for setup and connects messages by calculating the signature over the siie fields specified by the smme protocol it is still offered for any other signaling messages release

status restart and for setup and connect messages if the smme protocol is not used by introducing a signature calculated over part of the siie availability this goal means maintaining and assuring the accuracy and consistency of data the function of Integrity is to make sure that the data is reliable and is not changed by unauthorized persons so ATM security system should provide the following functions verification of identities security system should be able to establish and verify the claimed identity of any actor in an ATM network controlled access and authorization the actors should not be able to gain access to information or resources if they are not authorized to protection of confidentiality stored and communicated

data should be confidential protection of data Integrity the security system should guarantee the Integrity of the stored and communicated data strong accountability an entity cannot deny the responsibility of its performed actions as well as their effects activities logging the security system should support the capability to retrieve information about security activities in the network elements with the possibility of tracing this information to individuals or entities alarm reporting the security system should be able to generate alarm notification about certain adjustable and selective security related events audit when violations of security happen the system should be able to analyze the logged data relevant to security security recovery the security system

should be able to recover from successful or attempted breaches of security Security Management the security system should be able to manage the security services derived from the above requirements they are necessary to support the maintenance of Security Services if the security system cannot be recovered from attacks and not providing Security Services anymore then the system won't be secure after these attacks on the other hand Security Services and information about security have to be managed securely they are foundations of the security system threads to ATM as other networks ATM networks will suffer a lot of

threats typical ones are eavesdropping spoofing service denial VC stealing and traffic analysis

Etc notice that VC stealing and traffic analysis happen only in ATM networks eavesdropping eavesdropping refers to the threat that the attacker connects or taps into the transmission media and gain unauthorized access to data it is one of the most common attacks to the network since most ATM networks are connected with fiber optic cables some people might get the wrong impression that it is not so easy to tap a ATM network spoofing spoofing attack means that an attacker tries to impersonate another user to the third part therefore can get access to resources belonging to the victim to take advantage or just destroy them spoofing might need special tools to manipulate the protocol data unit sometimes it might require the attacker

has special access permission C must be super user in Unix environment however since a network will be connected to many interested networks via the Internet it's impossible to prevent a hacker from getting this excess permission or even trace the people with this particular excess permission ATM is being implemented in public domain therefore it is subjected to this kind of attack also service denial ATM is a connection oriented technique a connection which is called virtual circuit VC in ATM is managed by a set of 14 minutes, 1 second signals VC is established by setup signals and can be disconnected by release or drop party signals if an attacker sends release or drop party signal to any intermediate switch on the way of a VC then the VC will be

disconnected by sending these signals frequently the attacker can greatly disturb the communication between one user to another therefore will disable the quality of service in ATM combining this technique with other tricks like eavesdropping the attacker can even completely block one user from another stealing of VCS if two switches in an ATM network compromise the attacker can even steal a VC from another user say vc1 and vc2 are two virtual channels which will go through switch A and switch B where one is owned by user U1 and vc2 is owned by user U2 if A and B have compromised then A can switch VC one's cell going from A to B through vc2 and B will switch back those cells to vc1 since switches will forward cells based on the VCI that is virtual Channel identifier or VPI that is

virtual path identifier in the cell header A and B can just alter these fields back and forth switches between A and B won't notice these changes and will switch the assumed vc2's sale just like the authentic vc2's sales in public packet switching Network U1 won't gain too much by District however in an ATM network if quality of service is guaranteed then user one can gain a lot of stealing a higher quality Channel which user one is not entitled to use ACC to the excess control policy user one can gain even more if every user has to pay for the communication in both cases user two will be hurt someone may argue that the possibility that the switches will compromise is pretty low that will be true if the ATM network is owned by one

organization when we consider ATM inter networking in which Kells will travel through different ATM networks it will be very easy for two switches to compromise traffic analysis traffic analysis refers to a threat that the hacker can get information by collecting and analyzing the information like the volume timing and communication parties of a VC volume and timing can reveal a lot of information to the hacker even though the data is encrypted because encryption won't affect the volume and timing of information and also the source and this destination parties can be obtained from the cell header normally is in clear text and some knowledge about the routing table another related thre is called

convert channels in this technique the attacker can encode the information in the timing and volume of data VCI or even session key to release information to another people without being monitored normally these two attack won't happen however when ATM is used in an environment requiring stringent security it might happen traffic Management in order for ATM networks to deliver guaranteed quality of service on demand while maximizing the utilization of available network resources effective traffic management mechanisms are needed almost every aspect of ATM network operation from signaling requests and routing to network resource allocation and policing contains some traffic management mechanisms some of parameters are as follows quality of service attributes

while setting up a connection on ATM networks users can specify the following parameters related to the desired quality of service Peak cell rate PCR the maximum instantaneous rate at which the user will transmit for bursty traffic the inter cell interval and the cell rate varies considerably the PCR is the inverse of the minimum inter cell interval sustained cell rate ISC this is the average rate as measured along a long time interval cell loss ratio CLR the percentage of cells that are lost in the network because of error or congestion and are not delivered to the destination recall that each ATM cell has a cell loss priority the CLP bit in the header during periods of congestion the network will first discard cells with CLP is equal to 1 since the loss of cells with

CLP is equal to zero is more harmful to the operation of the application CLR can be specified separately for cells with CLP is equal to 1 and for those with CLP is equal to zero cell transfer delay ctd the delay experienced by a Cale between Network entry and exit points is called the sale transfer delay it includes propagation delays queing delays at various intermediate switches and service times at queing Point cell delay variation CDV this is mayor of variance of ctd high variation implies larger buffering for delay sensitive traffic such as Vo and video burst tolerance BT this determines the maximum burst size that can be sent at the peak rate this is the bucket size parameter for leaky bucket algorithm that is used to control the traffic entering the network the algorithm consist of putting all

arriving cells in a buffer bucket which is drained at the sustained cell rate is CR the maximum number of backto back cells that can be sent at the peak sell rate is called maximum burst size MBS minimum sell rate MCR this is the minimum rate desired by a user the first six of the

above traffic parameters were originally specified in a version 3 traffic contract to provide a guaranteed qos a traffic contract is established during connection setup which contains a connection traffic descriptor and a conformance definition however it is not necessary for every ATM virtual connection to have a specified qos the reason for this is that if only specified qos connections are supported by ATM then a large percentage of the network resources will be wasted this can happen when one or more connections

are not utilizing the full capacity of their qos contracts unspecified qos contracts can be supported by an ATM network on a best effort basis such best effort services are sufficient for supporting most of the existing data applications congestion control techniques congestion control lies at the heart of the general problem of traffic management for ATM networks in general congestion arises when the incoming traffic to a specified link is more than the outgoing link capacity the primary function of congestion control is to ensure good throughput and delay performance while maintaining a fair allocation of network resources to the user one method to avoid Network congestion is to accept a new ATM connection during connection setup phase

only when sufficient network resources are available to provide the acceptable qos this is called connection and Mission Control CAC which is needed for connections where the qos must be guaranteed disadvantages ATM has not been widely accepted although some phone companies still use it in their backbone networks the expense complexity and lack of interoperability with other technologies have prevented ATM from becoming more prevalent expense ATM technology provides a comprehensive list of services even a moderate ATM switch costs much more than inexpensive LAN Hardware in addition the network interface card needed to connect a computer to an ATM network is significantly more expensive than a

corresponding ethernet NIC connection setup latency ATM's connection oriented Paradigm introduces significant delay for distant communication the time required to set up and tear down the ATM VC for distant communication is significantly larger than the time required to use it sell tax ATM sell headers impose a 10% tax on all data transfer in case of ethernet sell taxes 1% lack of efficient broadcast connection oriented networks like ATM are sometimes called non-broadcast multiple access NB MA networks because the hardware does not support broadcast or multicast on an ATM network broadcast to a set of computers is simulated by arranging for an application program to pass a copy of data to each computer in the set as a result broadcast is inefficient complexity of

qos the complexity of this specification makes implementation cumbersome some and difficult many implementations do not support the full standard Assumption of homogeneity ATM is designed to be a single Universal networking system there is minimal provision for interoperating with other Technologies so friends this was all about ATM network security protocols hope the concepts explained in this lecture were understandable and helpful hope to see you in the next lecture till then goodbye thank you 25 minutes, 1 second [Music]